

- Q.5 An inverse transducer converts (CO1)
- a) Electrical energy into thermal energy
 - b) Electrical energy into light energy
 - c) Electrical quantity into mechanical quantity
 - d) Electrical quantity into a non electrical quantity
- Q.6 Amount of water vapour in air is regarded as (CO5)
- a) Water & steam b) Humidity
 - c) Mildness d) Pressure

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 What is Gauge Factor? (CO2)
- Q.8 Define Torque. (CO3)
- Q.9 What is a load cell? (CO3)
- Q.10 What is a thermistor? (CO5)
- Q.11 What is the unit of torque? (CO3)
- Q.12 What is measurement? (CO5)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain any method to measure speed of a motor. (CO3)

- Q.14 How torque can be measured explain any one method. (CO3)
- Q.15 Differentiate between primary transducer and secondary transducer. (CO1)
- Q.16 What are piezoelectric transducers? (CO1)
- Q.17 Explain the construction and working of semiconductor type of strain gauge. (CO2)
- Q.18 Explain working principle of any one type of pyrometer. (CO5)
- Q.19 How a potentiometer can be used as a displacement transducer. (CO1)
- Q.20 How you can use a capacitive transducer to measure level in a tank. (CO5)
- Q.21 Explain the working principle of a basic hair hygrometer. (CO5)
- Q.22 What is the significance of measurement in industrial field? (CO1)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Draw the block diagram of basic measurement system and explain each stage in detail. (CO1)
- Q.24 Explain in detail working principle, construction and applications of linear voltage differential transformer. (CO2)

Q.25 What do you mean by flow rate explain the working principle and construction of ultrasonic flow meter. List its advantages also. (CO4)